

EDUCATION	University of Florida	Gainesville, FL
	M.S. in Electrical and Computer Engineering GPA: 3.56/4.0	Aug. 2021 – May 2023
	Hubei University of Technology	Wuhan, China
	B.S. in Electrical Engineering GPA: 3.3/4.0	Jun. 2016 – Jun. 2020
PROFESSIONAL EXPERIENCE	University of Florida	Gainesville, FL
	<i>Data Scientist III</i>	Sep. 2025 – Present
	- Design and deploy advanced AI/ML models (deep learning, NLP, predictive analytics) on large-scale EHR datasets, improving patient safety and quality of care. - Lead cross-functional teams to develop scalable data pipelines and optimize feature engineering for clinical integration. - Deliver reproducible solutions with CI/CD, containerized deployment, and robust model monitoring, ensuring HIPAA compliance.	
	<i>Application Programmer</i>	Mar. 2024 – May 2025
	- Implemented MLOps workflows with CI/CD and monitoring, enabling scalable deployment on Linux systems. - Developed scalable ETL pipelines and engineered clinical features using cluster computing, boosting ML model performance on large-scale healthcare data. - Built and deployed secure, interactive healthcare web applications with FHIR integration on AWS, enhancing scalability, reliability, and clinical interoperability.	
	<i>Research Assistant</i>	Dec. 2022 – Feb. 2024
	- Cleaned and preprocessed extensive healthcare datasets, generating insightful visualizations to support data-driven research. - Developed a computer vision-based object detection system integrated with automated data acquisition pipelines, enabling efficient data collection and analysis. - Built and deployed interactive web platform for 3D modeling and predictive analytics, enhancing visualization of object detection outcomes and improving user experience.	
	Regenstrief Institute	Indianapolis, IN
	<i>Application Programmer</i>	Jun. 2025 – Aug. 2025
	- Standardized U.S. EHR data by architecting pipelines for OMOP and PCORnet models, improving data quality and harmonization. - Engineered automated, scalable ETL pipelines and extracted longitudinal clinical features from raw EHR data using distributed cluster computing. - Created web-based analytics dashboards to support clinicians in interpreting complex data and improving decision-making.	

- PUBLICATIONS
1. Yu Huang, Jie Xu, Zhengkang Fan, [Yu Hu](#), et al. “Identifying Alzheimer’s Disease Progression Subphenotypes via a Graph-based Framework using Electronic Health Records.” J Healthc Inform Res (2026).
 2. Jin Wang, [Yu Hu](#), et al. “Technical note: ShinyAnimalCV: open-source cloud-based web application for object detection, segmentation, and three-dimensional visualization of animals using computer vision.” Journal of Animal Science, vol. 102, 2024, skad416.

PROJECTS

Neonatal Lactation and Milk Intake Data Processing Project

Sep. 2025 – Present

- Processed and standardized raw lactation and neonatal milk intake data from the EHR, including harmonizing pumping records, feeding documentation, and inconsistent data entries.
- Worked closely with neonatologists and lactation specialists to define and validate **clinically meaningful lactation and feeding features**, such as NPO patterns, respiratory support status, maternal milk availability, feeding intolerance, and early nutrition indicators.
- Developed automated feature engineering pipelines and integrated machine learning models to support early identification of infants at risk for inadequate milk intake or feeding complications.
- Deployed model pipelines and data processing workflows onto on-premise servers, enabling reproducible analytics and continuous data updates.
- Built and connected **Power BI** dashboards to model outputs and clinical features, providing real-time visualization for clinicians and lactation teams to support data-driven feeding management.

Graph-Based Modeling for Alzheimer’s Disease Progression Prediction

Dec. 2023 – Nov. 2025

- Construct a patient similarity graph from **Electronic Health Records (EHR)** using **K-Nearest Neighbors (KNN)** to identify the top 100 most similar patient encounters.
- Utilize **NetworkX** to build the graph structure and integrate it with a **Graph Neural Network (GNN)** for predictive analysis of Alzheimer’s Disease progression.
- Developed deep learning models using **PyTorch** to enhance predictive algorithms. Optimized model training and inference speed using **GPU CUDA** programming.
- Identify disease progression subphenotypes to improve early detection of Alzheimer’s Disease and support personalized treatment strategies.

Advanced EHR Continuity Integration for Robust T2D Risk Prediction

Feb. 2024 – Nov. 2025

- Engineered an automated, **PCRONet**-based pipeline to systematically extract disease phenotypes, integrate longitudinal patient encounters, and execute comprehensive data cleaning on linked EHR and claims datasets from the **OneFlorida+ Clinical Research Network**.
- Developed and validated a machine learning framework for **type 2 diabetes** risk prediction, leveraging the **Harmonized Encounter Proportion Score (HEPS)** to quantify EHR data-continuity and improve predictive accuracy.
- Conducted fairness assessments using advanced algorithmic fairness metrics to identify disparities in model performance across racial-ethnic groups, ensuring equitable risk prediction.
- Utilized **cluster computing** and **parallel computation** techniques to optimize the system’s computational efficiency, significantly reducing processing time for large-scale datasets.

OCT-Driven Retinal Image Biomarker Analysis for ADRD Prediction

Dec. 2023 – Sep. 2025

- Develop an automated pipeline to process **OCT DICOM scans**, enabling precise extraction and quantification of retinal imaging data.
- Construct predictive models using retinal thickness metrics from OCT scans to assess **ADRD onset risk**, establishing a predictive link between ocular biomarkers and neurodegenerative progression.
- Integrate OCT-derived features into **machine learning frameworks** to evaluate the predictive relationship between retinal measurements and ADRD development, enhancing early detection capabilities.
- Utilize **Large Language Models (LLMs)** to extract critical information from notes data, improving the accuracy and comprehensiveness of the analysis.

Healthcare Risk Prediction Web Application

Nov. 2023 – Sep. 2025

- Engineer front-end components using **React.js**, implementing secure user authentication and patient selection interfaces to enhance user experience.
- Develop a back-end **Flask** API for model integration, leveraging causal inference models to assess patient risks within the prediction framework.
- Utilize **SQL** to query and extract relevant data from healthcare databases, ensuring accurate and timely information retrieval.
- Integrate **FHIR-based database** and **EPIC system** inquiry functionality to streamline patient data retrieval and improve interoperability with electronic health records, while containerizing and orchestrating the entire software stack using **Docker**.

OMOP-FHIR Database Integration and Deployment Platform

Apr. 2024 – Sep. 2025

- Design and develop a comprehensive **OMOP database** to support standardized healthcare data integration and advanced analytics.
- Deploy and evaluate a web-based platform integrating **OMOP** and **FHIR** query capabilities, enhancing data interoperability and enabling real-time access.
- Utilize **ontology concept codes** to query, debug, and maintain the database, ensuring data consistency and accuracy.

AWARDS
AND
HONORS

- **Honor:** Herbert Wertheim College of Engineering Achievement Award 2021
Scholarship (Awarded for academic excellence and research contributions)
- **Scholarship:** Undergraduate Scholarship awarded to top 5% of students 2020
for academic excellence in Electrical Engineering.

SKILLS

Programming Languages: Python, SQL, R, C/C++, Java, JavaScript, HTML, CSS, LaTeX, Shell, Matlab

Machine Learning & Data Analysis: PyTorch, TensorFlow, Scikit-learn, XGBoost, NetworkX, Causal Inference

Tools & Platforms: Docker, Linux, Tableau, Power BI, Cluster Computing, Cloud Computing, Git

ACADEMIC
SERVICES

Reviewers for: *IEEE-ICHI*